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RESEARCH ARTICLE

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Beyond stroke: atrial fibrillation and the broader burden of thromboembolic events in the population-based EpiHealth study

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ABSTRACT

Objectives: To investigate the distribution and risk of thromboembolic events (TEs) among adults with and without atrial fibrillation (AF), and to assess oral anticoagulant (OAC) use over time. **Design:** This population-based longitudinal cohort study included EpiHealth participants recruited between April 2011 and December 2016 and followed through linkage to Swedish health data registers from baseline until the first TE, death, or December 31, 2016. **Results:** At baseline, 873 of 22,616 participants (3.9%) had AF and 297 developed incident AF during follow-up. 484 participants experienced at least one TE during 70,587 person-years. Participants with AF had higher TE incidence (68 events, 5.9%; $p < 0.001$), predominantly arterial (73.5%), with 94% of arterial events involving the cerebral circulation. Approximately one quarter of TEs among participants with AF occurred outside the cerebral circulation and comprised both venous and non-cerebral arterial events. Although participants with incident AF represented less than one quarter of participants with AF, they accounted for 40% of TEs in the AF group. OAC use increased from 39.0% to 50.4%, with a shift from predominantly warfarin to direct oral anticoagulants. In time-dependent Cox regression adjusted for age, sex, and prior arterial thromboembolism, untreated AF was associated with increased thromboembolic risk compared with no AF (HR 2.53), whereas OAC use was associated with lower risk. **Conclusions:** AF was associated with substantially increased thromboembolic risk, predominantly arterial in nature. One in five TEs among participants with AF coincided with newly diagnosed AF, suggesting potential opportunities for earlier AF detection and timely initiation of anticoagulant treatment.

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Atrial fibrillation; arterial thromboembolism; venous thromboembolism; oral anticoagulants; risk reduction

Introduction

Atrial fibrillation (AF) is associated with risk of complications including thromboembolism [1]. The association between AF and ischemic stroke is well-established [2] and about 25% of ischaemic strokes can be attributed cardiogenic emboli caused by AF [3]. It is common that AF is undiagnosed before the embolic stroke [3]. Medical guidelines for ischemic stroke care include screening for undetected AF [4] and to prevent future ischemic strokes by prescribing oral anticoagulants (OAC). Methods used for risk stratification of stroke include the CHA_2DS_2-VA score recommended by European guidelines [1].

In recent decades, OAC therapy has advanced significantly. Since the introduction of direct oral anticoagulants (DOAC) in the 2010s [5], prescription patterns have shifted [6], and today DOACs are the preferred treatment for preventing thromboembolism in patients with AF [1].

Beyond cerebral infarction, AF may also lead to non-cerebral arterial thromboembolism (ATE), affecting peripheral arteries or visceral organs [7–9]. Previous studies have shown that AF-associated cardioembolic events accounts for 60–95% of acute limb ischemia, 31% of splenic infarction, 55% of renal infarction and 47% of acute mesenteric ischemia [10].

Besides its well-established impact on the arterial vascular system, AF is also associated with venous thromboembolism (VTE) including pulmonary embolism (PE) [11–13], although the direction of causality remains debated. PE can increase pulmonary pressure and precipitate right heart failure, which in turn may

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trigger AF. Conversely, AF may theoretically promote thrombus formation in the right atrium, potentially leading to PE although this pathway has been less extensively studied.

Despite increasing interest in AF and thromboembolism, the overall distribution of thromboembolic events (TEs), including stroke, non-cerebral ATE, and VTE, remains incompletely characterized among individuals with AF in the general population.

Accordingly, the primary aim of this study was to describe the occurrence and distribution of ATE and VTE among participants with and without AF in the EpiHealth cohort. Secondary aims were to assess impact of anticoagulant treatment use as well as to compare thromboembolic risk between participants with and without AF.

Material and methods

Study design and population

This longitudinal study followed participants through national registries until the first TE, death, or end of follow-up.

The study was based on the EpiHealth cohort, a large Swedish population-based study. Equal numbers of men and women aged 45–75 years, stratified by five-year age groups, were randomly selected from the general population through the Swedish population register in the municipalities of Uppsala and Malmö and invited to participate.

Between 2011 and 2018, 100,685 individuals were invited, of whom 25,369 (25.5%) completed the questionnaire. Further details of the EpiHealth project have been published previously [14].

At the baseline visit, participants completed a comprehensive web-based health questionnaire and underwent physical measurements, including electrocardiogram (ECG), all assessed by specialist physicians in angiology or cardiology. A bipolar single-lead extremity ECG (Zenicor, Stockholm, Sweden) [15] was used for the duration of 30 s between the participant's hands and ECG electrodes to screen for AF. AF was defined as one 30-second recording with irregular rhythm without p waves [16].

Participants were included if they took part in EpiHealth between April 2011 and December 2016. Exclusion criteria were missing or invalid ECG recordings after physician validation and age <45 years at participation. Participants younger than 45 years were excluded, as they were outside the intended age range of the study population.

Ethics

The principles of the Declaration of Helsinki were followed with informed written consent. The EpiHealth study was approved by the Ethics Board in Uppsala (Dnr 2010/402) and the Ethics Board in Lund (Dnr 2016/945).

Definitions of terms and outcomes

Atrial fibrillation and atrial flutter, AF

Atrial fibrillation and atrial flutter were not differentiated, as they share identical anticoagulation recommendations [1]; AF refers to both.

Participants were classified according to AF status at baseline. The AF group included screening-detected AF, prior diagnoses, or self-reported AF. Participants diagnosed with AF during follow-up were initially classified as non-AF and reclassified at the date of diagnosis.

Definitions of outcomes (thromboembolic event, time of death or end of study period)

The study cohort was linked with the Swedish national patient register (SNDR) administered by the National Board of Health and Welfare (www.socialstyrelsen.se/english) adding nationwide data on diagnosis codes from in- and specialised outpatient visits from 1987. Additional data from theprescribed drug register and the cause of death register as well as data from the quality register he Swedish stroke register were linked to the cohort.

Subject were followed from the EpiHealth baseline visit until December 31, 2016, death, or censoring. The primary outcome was incident TE, defined as a first registered event without prior documentation of the same diagnosis. International Statistical Classification of Diseases and Related Health Problems, Tenth Revision (ICD-10) codes included arterial thromboembolism (I63, G45.9, G45.3, I74, N28.0, K55.0, D73.5) and venous thromboembolism (I26, I80.1–I80.9, I82.2–I82.3, I82.8–I82.9). Both primary and secondary diagnoses were considered.

CHA_2DS_2-VASc score and OAC treatment

The CHA_2DS_2-VASc score, which guided anticoagulation at the time of data collection, was calculated for all participants without AF and for those with known AF at baseline [17]. Comorbidities were identified using ICD-10 codes for heart failure (I50–I50.9), hypertension (I10–I13, I15), prior stroke, transient

ischemic attack (TIA), or ATE (I63, G45.9, I74), vascular disease (I21, I22, I25, I70), and diabetes mellitus (E10–E14). OAC treatment among participants with AF was identified from the prescribed drug register. OAC use was assessed at baseline and at the time of any TE.

At the time of the study, OAC was recommended for patients with AF and CHA_2DS_2-VASc ≥2 in men and ≥3 in women. DOAC treatment was considered ongoing if dispensed prescriptions matched defined daily doses, while warfarin use was inferred from repeated dispensations.

Statistical analysis

To describe the occurrence and distribution of TE, as well as the use of OACs, descriptive statistics were applied.

The association between AF and TE was assessed using a Cox proportional hazards regression model adjusted for age, sex, previous ATE, and OAC treatment Variables included in the multivariable Cox regression model were prespecified based on clinical relevance and their potential role as confounders of the association between AF and TE.

To assess whether OAC treatment modified the association between AF and TE, an interaction term between AF and OAC (Warfarin and DOAC) category was included.

AF status was modeled as a time-varying covariate using a start–stop format, allowing participants who developed AF during follow-up to change exposure status at the time of diagnosis. Participants with AF and TE diagnosed on the same date (considered likely undetected or pre-existing AF) were excluded from the Cox models. Hazard ratios (HRs) with 95% confidence intervals (CIs) were reported. A sensitivity analysis was conducted by repeating the Cox regression model as two separate models, with ATE and VTE as the respective outcomes. Analyses were conducted using IBM SPSS Statistics version 29 (IBM Corp., Somers, NY), and Cox regression analyses were performed in R (using the *survival* package).

Results

At baseline, 873 out of 22,616 participants (3.9%) had AF. During follow-up through December 31, 2016, 297 participants were diagnosed with incident AF (incidence rate 4.38 [95% CI 3.90–4.91] per 1,000 person-years). Twelve incident AF diagnoses occurred after a TE; for analyses of thromboembolic outcomes, these individuals were classified as non-AF.

Figure 1 shows number of participants in study, with and without AF at baseline and during follow-up. Baseline characteristics (including CHA_2DS_2-VASc) for the groups no AF and AF are presented in Table 1.

During follow-up, 484 participants experienced at least one TE over 70,587.4 person-years, corresponding to an incidence rate of 6.86 per 1,000 person-years (95% CI, 6.26–7.50). Most TEs were arterial (248 events, 51.2%), with the brain being the most frequently affected organ regardless of AF status; renal, splenic, and visceral events were rare.

Among participants without AF, 416 individuals experienced TE, most commonly in the venous circulation (218 events, 52.4%). Participants with AF had a significantly higher TE incidence (68 events, 5.9%; $p < 0.001$), predominantly arterial (50 events, 73.5%), with most involving the cerebral circulation.

During follow-up, 297 participants were diagnosed with AF, of whom 285 were diagnosed before or at the time of TE. Although this subgroup represented <25% of all participants with AF ($n = 1{,}158$), it accounted for 40% of TEs within the AF group (27/68). Fourteen participants (7 women; mean age 68.2 ± 5.8 years) were diagnosed with AF and TE on the same date, including ten strokes/TIAs, three PEs, and one splenic infarction.

Distribution of TEs is provided in Table 2.

At baseline, 795 participants had known AF. The mean CHA_2DS_2-VASc score was 1.73 ± 1.26, and 39% (311/795) received OAC treatment, predominantly warfarin (79%), with 21% treated with a DOAC. By the end of follow-up, the mean CHA_2DS_2-VASc score had increased to 1.85 ± 1.34, and OAC use rose to 50.4% (401/795), with a shift toward DOACs (52% DOAC, 48% warfarin). Baseline CHA_2DS_2-VASc scores and OAC use are presented in Table 3.

At the individual level, 293 with AF participants remained on OAC throughout follow-up (173 on warfarin, 70 on DOAC, and 50 switched from warfarin to DOAC), 108 initiated OAC during follow-up (20 warfarin, 88 DOAC), 18 discontinued their OAC treatment, and 348 remained untreated. Among participants with incident AF, 74% were prescribed OAC, of whom approximately three-quarters received a DOAC.

In a multivariable time-dependent Cox proportional hazards regression analysis adjusted for age, sex, and previous ATE, AF without OAC treatment was associated with a higher risk of TE compared with individuals without AF and not receiving OAC (HR 2.53). This association was substantially attenuated among participants with AF who received DOAC therapy, corresponding to an estimated hazard ratio of 1.75, and was further reduced among those treated

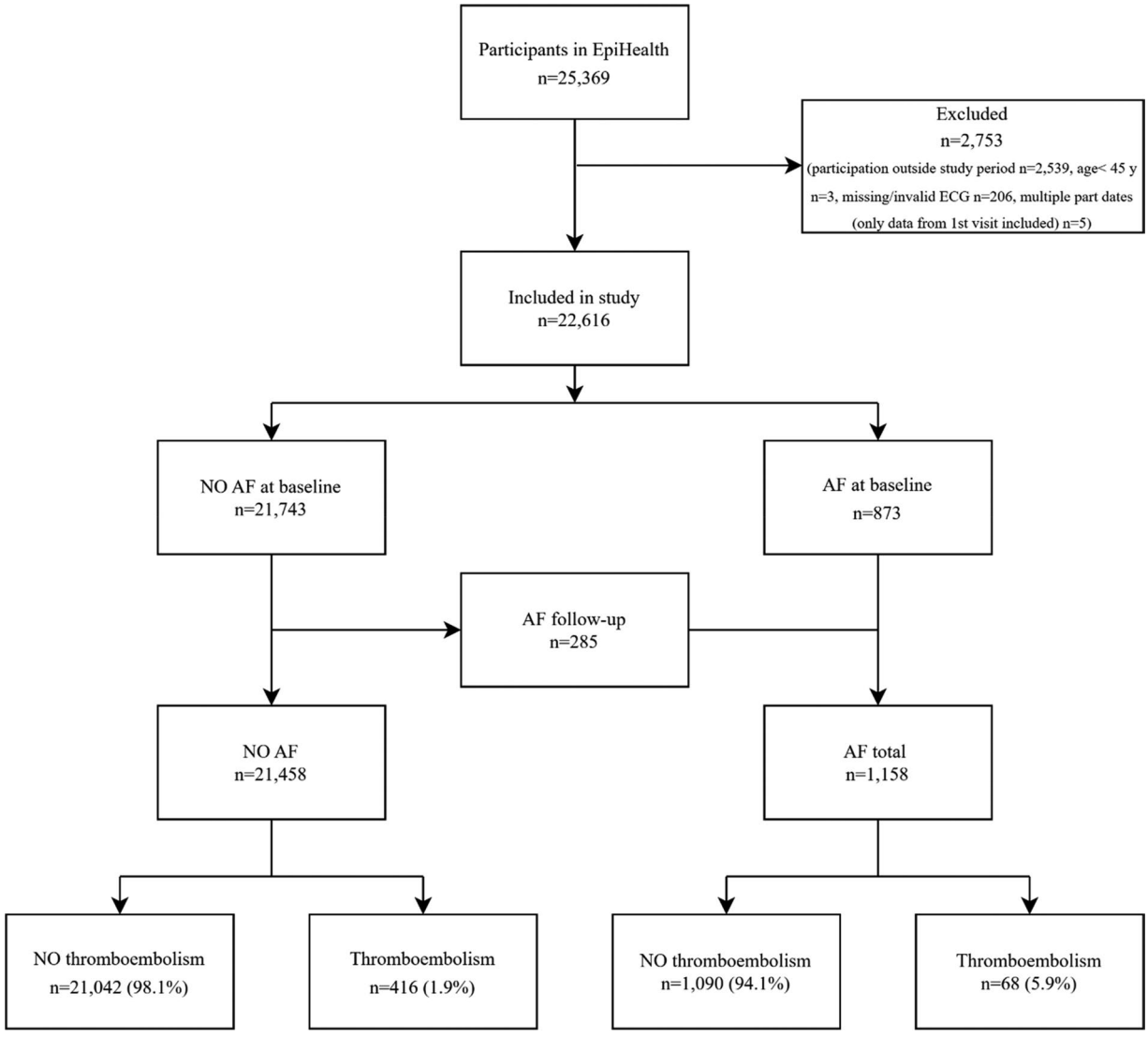


Figure 1. Flowchart of participants from inclusion to end-of-study. AF, atrial fibrillation.

Table 1. Baseline characteristics.

	No AF ($n=21,743$)	AF at baseline ($n=873$)
Age, years (SD)	60.12 (8.5)	66.25 (6.8)
Female, *n* (%)	12,432 (57.2)	286 (32.8)
Smoking, *n* (%)[a]		
Non-smoker	9,386 (43.1)	347 (39.7)
Past/Present smoker	9,795 (45.0)	456 (52.2)
Hypertension, *n* (%)	5,925 (27.3)	505 (57.8)
Congestive heart failure, *n* (%)	166 (0.8)	139 (15.9)
Ischemic heart disease, *n* (%)	897 (4.1)	161 (18.4)
CHA_2DS_2-VASc (SD)[b]	1.12 (0.9)	1.73 (1.3)

[a]Missing data on smoking were present for 2,562 participants in the no AF group and 70 participants in the AF group.
[b]CHA_2DS_2-VASc was calculated by using registry data. Other variables consist of data from health questionnaire and diagnose register.
AF, atrial fibrillation; SD, standard deviation; CHA_2DS_2-VASc, congestive heart failure, hypertension, age ≥75 years, diabetes mellitus, stroke/transient ischemic attack/thromboembolism, vascular disease, age 65–74 years, and sex category.

Table 2. Distribution of thromboembolic events by atrial fibrillation status.

Event type	No AF, *n* (%)	AF, *n* (%)	Total events, *n*
Arterial thromboembolic events	198 (47.6)	50 (73.5)	248
Stroke/TIA	184 (44.2)	47 (69.1)	231
Renal infarction	2 (0.5)	0 (0.0)	2
Splenic infarction	0 (0.0)	1 (1.5)	1
Visceral embolism	2 (0.5)	0 (0.0)	2
Extremity embolism	10 (2.4)	2 (2.9)	12
Venous thromboembolic events	218 (52.4)	18 (26.5)	236
Pulmonary embolism	101 (24.3)	10 (14.7)	111
Deep vein thrombosis	117 (28.1)	8 (11.8)	125
Total thromboembolic events	416 (100)	68 (100)	484

AF, atrial fibrillation; TIA, transient ischemic attack.
Note: Percentages are calculated using the total number of thromboembolic events within each AF group as the denominator (no AF: 415 events; AF: 69 events).

Table 3. CHA_2DS_2-VASc score and OAC treatment among participants with AF prior to baseline and at the end of study, stratified by sex.

CHA_2DS_2-VASc score	Men $n=529$ (66.5%)	Women $n=266$ (33.5%)	Men OAC n (%)	Women OAC n (%)
Baseline				
0	114 (21.6)	—	10 (8.8)	—
1	197 (37.2)	79 (29.7)	88 (44.7)	5 (7.0)
2	115 (21.7)	110 (41.4)	66 (57.4)	29 (27.9)
3	58 (11.0)	44 (16.5)	34 (58.6)	28 (49.1)
>4	45 (8.5)	33 (12.4)	33 (73.3)	20 (60.6)
End-of-study				
0	108 (20.4)	—	20 (18.5)	—
1	180 (34.0)	75 (28.2)	92 (51.1)	12 (16.0)
2	123 (23.3)	102 (38.3)	89 (72.4)	37 (36.3)
3	64 (12.1)	47 (17.7)	43(67.2)	28 (60.0)
>4	54 (10.2)	42 (15.8)	40 74.1)	32 (76.2)

During 2011–2016, OAC treatment in patients with AF was considered for CHA_2DS_2-VASc 1 in men or 2 in women and was recommended for CHA_2DS_2-VASc ≥2 in men or ≥3 in women.
AF, atrial fibrillation; OAC, oral anticoagulant; TE, thromboembolic event; CHA_2DS_2-VASc, congestive heart failure, hypertension, age ≥75 years, diabetes mellitus, stroke/transient ischemic attack/thromboembolism, vascular disease, age 65–74 years, and sex category.

Table 4. Cox proportional hazards model evaluating the association between AF and TEs, including interaction effects with anticoagulant therapy.

Main effects	Hazard ratio	95% CI	p-value
AF (ref: no AF)	2.53	1.77–3.63	<0.001
Age (per year)	1.06	1.05–1.08	<0.001
Male sex	1.52	1.26–1.83	<0.001
Previous ATE	3.11	2.14–4.52	<0.001
Interaction terms			
AF × direct OAC	0.14	0.05–0.39	<0.001
AF × Warfarin	0.13	0.06–0.32	<0.001

Model adjusted for age, sex, and previous ATE.
CI, confidence interval; AF, atrial fibrillation; TE, thromboembolic event; OAC, oral anticoagulant; ATE, arterial thromboembolism.
Hazard ratios are from a multivariable Cox proportional hazards model adjusted for age, sex, and previous ATE. The reference category for anticoagulant treatment is no oral anticoagulant use.

with warfarin (HR 1.14). Results of cox regression and interaction effects are summarized in Table 4.

Sensitivity analyses excluding VTE and ATE separately produced similar estimates (if AF HR of ATE = 2.88 (95% CI: 1.8–4.59, $p<0.001$), (if AF HR of VTE = 2.16 (95% CI: 1.24–3.75, p 0.006).

Discussion

In this population-based study, AF was associated with a higher risk of TE than in individuals without AF. TE distribution differed by AF status: ATEs and VTEs occurred at similar rates among those without AF, whereas ATEs predominated in individuals with AF. During follow-up, OAC use increased, with a shift from warfarin to DOACs. This study extends previous knowledge by characterizing the broader spectrum of TEs associated with AF in a population-based cohort. In addition, the study period coincided with the transition from warfarin to DOACs, providing insight into changing anticoagulant treatment patterns.

In line with previous evidence, AF was associated with a higher risk of ATE. Cerebral embolism was the predominant manifestation, while approximately 6% of ATE were non-cerebral. Our findings are lower compared to those found by Bekwelem et al. (non-cerebral ATE accounted for 11.5% of all ATE) [9] and Orgel et al (8.2% of ATE were outside of cerebral circulation) [18]. Possible explanations for the slightly different results may include differences in population characteristics, including difference in age as well as a lower overall vascular risk burden.

Male sex was independently associated with a higher risk of TE, which may partly reflect sex differences in cardiovascular risk profiles and comorbidity burden previously observed in the EpiHealth cohort [19].

The predominance of cerebral embolism is consistent with established AF-related mechanisms, whereby thrombi formed in the left atrium preferentially embolize to the cerebrovascular circulation due to arterial anatomy and hemodynamic flow patterns. Previous studies have shown that patients with cerebral and non-cerebral ATE have similar age and CHA_2DS_2-VASc profiles, although non-cerebral ATE appears more common among women and smokers [9]. Later findings also link female sex, and in some studies higher CHA_2DS_2-VASc scores, to non-cerebral ATE [7]. In our study there were fewer women in the AF group, potentially biasing results in regard to ATE. Using an older cohort, representing both sexes equally might affect the percentage of ATE.

Lastly, as symptoms of non-cerebral ATE are often less specific than those of stroke, these conditions are most likely underdiagnosed. Both autopsy studies and clinical studies using more extensive imaging have identified a higher burden of TEs than detected in routine clinical care [8,20].

Previous research has suggested that AF is associated with an increased risk of both ATE and VTE, particularly shortly after AF diagnosis [13]. While a Swedish registry study [21] have reported an increased risk of PE early after AF diagnosis, others have not supported AF as an independent predictor of PE [22]. Recent results from Liu et al. similarly found no convincing evidence for a causal role of AF in PE development [23]. In our study, ATE was more common overall, however, VTE still accounted for nearly one quarter of TEs among individuals with AF. A few cases of PE were diagnosed with PE and AF simultaneously. However, the data did not allow for any causal conclusion to be made.

Although OAC therapy indicated a reduced risk of VTE among those with AF this risk was not fully

eliminated. Persistent VTE-related risk factors, particularly malignancy, remain important, and clinical considerations such as frailty, comorbidities, or bleeding risk may preclude OAC use despite guideline-based indications. In addition, treatment decisions ultimately depend on patient preference. Lack of compliance or other factors contributing to failure to achieve treatment goals may also have played a role. However, data on reasons for OAC initiation or non-use were not available in this study. The apparently stronger protective association observed with warfarin treatment compared with DOACs should be interpreted with caution, as the study was not designed to compare the treatments. The observed difference may partly reflect variations in treatment indications, patient characteristics, and comorbidities between treatment groups.

The burden of AF is increasing, with a lifetime risk estimated at 1 in 4 in men and women over 40 years of age [24]. In our cohort, the incidence of AF was 4.38 per 1,000 person years. The lower observed incidence may be explained by the relatively young mean age, short follow-up, and the fact that a substantial proportion of AF cases in Sweden are diagnosed and managed in primary care and were therefore not captured in this study. Consequently, the true incidence of AF in the general population is likely higher than observed.

The risk of TE was higher in participants with AF, consistent with previous findings. As described by Wieloch et al. Sweden has had a well-functioning warfarin treatment infrastructure and a follow-up of treatment *via* a national quality register since 2006 [25]. This has resulted in high warfarin use and high numbers of treatment within treatment recommendation in Sweden. This combination might have influenced our results. All of the currently used DOACs were introduced to the Swedish market during the study follow-up [26]. Although DOACs were initially prescribed primarily for AF-related anticoagulation, their use has since expanded. Their ease of use compared to warfarin has facilitated long-term anticoagulation, including extended treatment after VTE, which may also reduce the risk of other TEs irrespective of AF status.

Strengths and limitations

EpiHealth large cohort study which enables us to inspect associations in cardiac conditions, however true prevalence rates should be taken cautiously as participation rate was fairly low. In addition, participants in EpiHealth have been shown to have a higher educational level compared to the general Swedish population, likely contributing to healthy participant effect, with might further affect the generalisability of the results. As the primary objective of this study was to characterize the distribution of TE rather than to estimate the prevalence of AF, the relatively low participation rate is expected to primarily affect statistical power. There is limited evidence to suggest that the participation rate would have resulted in differential misclassification of either AF or TE.

A limitation to the study is that we had no data from primary care register, however by adding self-reported data to register data we aimed to get more accurate description of comorbidities, where many of the diseases may only be treated in primary care. Using register data from both in- and outpatient specialised care is a strength when investigating TEs, and we assume that all TEs (if diagnosed) were treated in specialised care and included in the register data.

The prescribed drug register includes all prescriptions dispensed however we have no data on actual use of the medication, whether correct dose was used, especially for warfarin, where correct dose is highly individual. Neither do we have data on reasons for any discontinuation the treatment.

Conclusion

In this population-based cohort study, AF was associated with a substantially higher risk of TE than in individuals without AF, with distinct arterial and venous patterns. ATE predominated among participants with AF, whereas ATE and VTE occurred at similar rates among those without AF. Non-cerebral ATEs constituted a small but consistent proportion of ATE irrespective of AF status. During follow-up, OAC use increased, with a shift from warfarin to DOACs. Notably, AF and TE were diagnosed simultaneously in one in five events among participants with AF, suggesting potential opportunities for earlier AF detection and timely initiation of anticoagulant treatment.

Clinically, these findings support efforts to improve early AF identification and highlight the importance of considering non-cerebral thromboembolic manifestations in patients with atypical presentations to avoid delayed diagnosis and treatment.

Author contributions

CRediT: **Pernilla Höiby**: Data curation, Formal analysis, Methodology, Visualization, Writing – original draft;

Moncef Zarrouk: Investigation, Methodology, Supervision, Writing – review & editing; **Lars Lind**: Conceptualization, Investigation, Writing – review & editing; **Rani Basna**: Formal analysis, Methodology, Writing – review & editing; **Marie Claire Overton**: Methodology, Supervision, Writing – review & editing; **Sölve Elmståhl**: Conceptualization, Funding acquisition, Methodology, Project administration, Supervision, Writing – review & editing.

Generative AI statement

During the preparation of this manuscript, ChatGPT (OpenAI, GPT-5.5) was used to assist with language editing and improve clarity. The authors taken full responsibility for the final content.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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